

Bilel Hatmi

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Applied mathematician (CentraleSupélec; Part III, Cambridge) working on the means to make AI more beneficial to society and less alienating. My interests combine the **psychometrics of AI models** with AI applied to **education and learning**, and to **conflict resolution** through game theory. At SRIE, I supervise three projects I designed on these questions.

RESEARCH

SRIE • Resident Researcher & Research Supervisor

Jul 2026 • Present

Eight-week AI-safety placements for Cambridge mathematics undergraduates, in progress

- **Seven undergraduates** (3/2/2) on three projects about what AI does to the people who use it, supervised through weekly group and one-to-one meetings.
- Organised a **one-week general training** on the conditions under which a trait measured in an LLM is valid.
- **Sycophancy as a set of constructs**: four separately caused quantities (direction, conviction, trajectory validity, dispersion), read by two instruments (behaviour; linear probes and persona vectors) across five conditions; design reviewed by Vardhan Kumar Ray (ex-Iliad, ex-SPAR).
- **Mediation under confidentiality constraints**: an LLM mediator whose neutrality is a property of the objective function; game-theoretic model, agentic workflow, synthetic-persona proof of concept.
- **An AI tutor against cognitive surrender**: knowledge-space model with a metacognitive component, LLM-persona construction of the domain, Bayesian state estimation, adaptive strategy by information gain.

MASt Essay: Proximal Causal Inference with Proxy Variables

2025 • 2026

Supervisors: Dr P. Zhao, Prof. Q. Zhao (DPMMS). Submitted May 2026.

Cambridge, UK

- Synthesis of estimators from linear to doubly-robust; **Python library** for practitioners estimating an average treatment effect nonparametrically under an unmeasured confounder.
- Simulations plus **two real-data external validations** (returns to schooling; arterial catheterisation and ICU mortality); binary policies smoothed by a noisy Gaussian kernel to stabilise the ill-posed inverse problem.

The Flattening • Finalist, Cambridge–McKinsey Risk Prize 2026

Mar 2026

Author & Developer

Cambridge, UK

- Essay on how AI productivity gains hide cognitive homogeneity; **Monte Carlo model** of systemic risk coupling workforce substitution, epistemic homogenisation and institutional resilience; interactive React application.

CartesIA • Founder & Lead Researcher

2025 • Present

LLMs as controlled measurement instruments for psychometrics

Paris & Cambridge

- **Protocol** mapping natural language onto validated constructs (agency, value–action alignment, autonomy, eudaimonic well-being), with contextualised vignettes, scoring rubrics and safeguards against suggestion bias.
- **Three domains** (mental health, youth guidance, professional development); diagnosis and surveillance excluded.

EDUCATION

University of Cambridge

2023 • 2026

MASt in Mathematical Statistics (Part III), DPMMS

Cambridge, UK

- Intermission for health; resumed early 2025; **degree completed June 2026**.
- Courses: Causal Inference · Robust Statistics · Modern Statistical Methods · Statistical Learning in Practice · Information Theory · Advanced Probability · Ethics in Mathematical Sciences

CentraleSupélec (Paris-Saclay)

2021 • 2026

Cursus Ingénieur (Grande École) • Major: Mathematics & Data Sciences

Paris, France

- **Top 1% of class** · GPA 4.28/4.33 (970 students) · Mathematical Modelling track (MMF) · degree completed 2026
- Courses: Advanced Probability & Statistics · Machine Learning · Game Theory · Optimisation · Philosophy of Science

Lycée Saint-Louis

2018 • 2021

MPSI / PSI* (Classes Préparatoires aux Grandes Écoles)

Paris, France

- Two-year intensive preparation for the national Grandes Écoles entrance exams.

PROFESSIONAL EXPERIENCE

Eleven Strategy <i>Analyst • Data Scientist (Strategy Consulting)</i>	<i>Apr 2025 • Sep 2025</i> <i>Paris, France</i>
• Buy-side due diligence on B2C targets: market sizing, competitor benchmarking, business plan support; recommendations used post-deal by senior management. • Prototyped AI agents (company research, CRM pre-processing) and an internal RAG system (embedding pipeline, metadata-aware indexing, lightweight evaluation), doubling retrieval precision for about 80 users.	
Elements Impact • “Boussole” <i>Research Intern</i>	<i>Jun 2023 • Sep 2023</i> <i>Paris, France</i>
• Operationalised subjective well-being indicators into predictive targets; trained models on a 20-year longitudinal panel (over 1,000 individuals); sensitivity and scenario analyses of measurement stability, under Emmanuelle Bioud (PhD, Cognitive Science).	
MICS Lab, CentraleSupélec <i>Research Assistant • Mathematical Modelling</i>	<i>Jan 2022 • Jul 2023</i> <i>Paris, France</i>
• Bayesian and stochastic modelling of myeloproliferative syndromes with the Gustave Roussy Institute, deriving a screening-age distribution under uncertainty; Hawkes process estimation (MLE, Whittle).	
CentraleSupélec <i>Teaching Assistant (Chargé de TD)</i>	<i>Sep 2022 • Jul 2023</i> <i>Paris, France</i>
• Probability, Integration & Convergence; Partial Differential Equations.	

SKILLS

Measurement	construct validity, reliability and generalisability analysis, psychometric operationalisation, experimental design for behavioural evaluations of LLMs
Causal inference	proximal causal inference, negative controls, doubly-robust and minimax estimation, ill-posed inverse problems and regularisation
LLM systems	agentic workflows, RAG design and evaluation, prompt engineering, Claude Code
Statistics & ML	supervised learning (GLM, XGBoost/RF), Bayesian modelling, stochastic processes (Hawkes), robust statistics, model diagnostics and calibration
Programming Languages	Python (pandas, numpy, scikit-learn, matplotlib), SQL, R (basic), LaTeX, Git French (native), English (C1), Arabic (fluent), Spanish (basic)

ADDITIONAL

Competitive	Jane Street R3 (Europe Trading Challenge); internship offer from QRT (quant fund).
Writing samples	The Flattening (essay); Part III essay on proximal causal inference; SRIE project briefs; Hawkes process technical report, available on request.
Citizenship	French & Tunisian.

REFERENCES

Dr P. Zhao / Prof. Q. Zhao	MASt essay supervisors, University of Cambridge, DPMMS
Emmanuelle Bioud	Research internship supervisor, Elements Impact (PhD, Cognitive Science)
Huey L.	Founder, SRIE